



GLOF database of High Mountain Asia

Glacier lake outburst floods (GLOFs) have been intensely investigated in High Mountain Asia (HMA) in recent years and are the most well-known hazard associated with the cryosphere. As glaciers recede and surrounding slopes become increasingly unstable, such events are expected to increase, although current evidence for an increase in events is ambiguous. Many studies have investigated individual events and while several regional inventories exist, they either do not cover all types of GLOF or are geographically constrained. Further, downstream impacts are rarely discussed. Previous inventories have relied on academic sources and have not been combined with existing inventories of glaciers and lakes. In this study, we present the first comprehensive inventory of GLOFs in HMA, including details on the time of their occurrence, processes of lake formation and drainage involved as well as downstream impacts. We document 766 individual GLOFs that occurred between 1533 and 2025. Of these, 23% were recurring events from just three ephemeral ice-dammed lakes. In combination, the documented events resulted in 8996 fatalities of which 906 fatalities were from 24 events, which is three times higher than a previous assessment for the region. The integration of previous inventories of glaciers and lakes within this database will inform future assessments of potential drivers of GLOFs, allowing more robust projections to be developed. The database and future, updated versions, are traceable, version controlled and can be directly incorporated into further analysis. This dataset has been updated in December 2025.

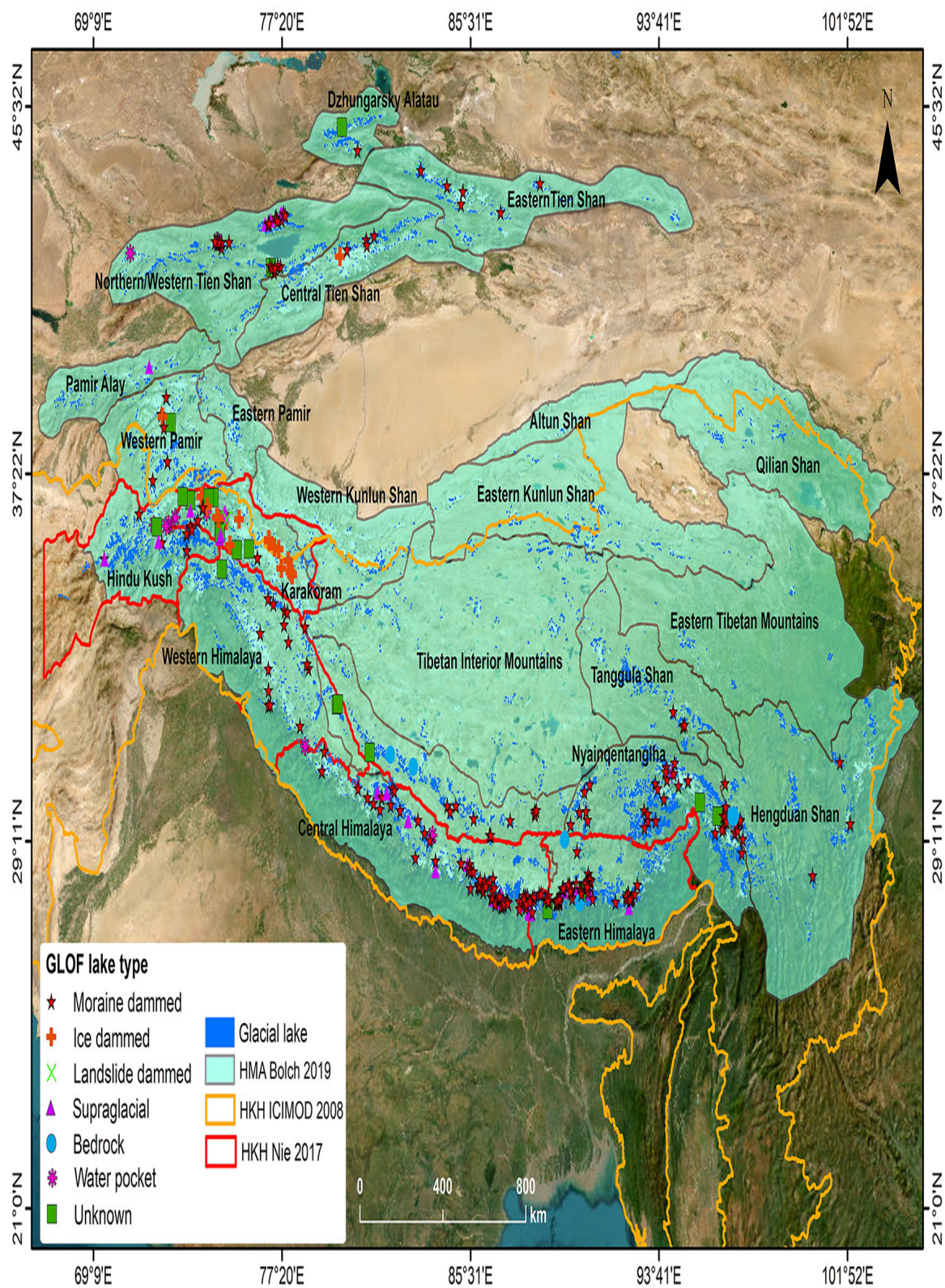
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Date (Publication)	2022-10-19
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Other citation details	ICIMOD. (2023). GLOF database of High Mountain Asia [Data set]. ICIMOD. https://doi.org/10.26066/RDS.1973283
Purpose	The database provides the first comprehensive dataset of GLOFs in the HMA, including their location, timing, associated processes, and downstream impacts. It includes a rigorous evaluation of local sources on previously unrecorded events to complement records from scientific literature. The dataset is fully accessible and interoperable to couple with other geospatial datasets to predict future projection of the hazards and potential drivers of GLOFs.
Status	Completed
Maintenance and update frequency	As needed
Keywords (Theme)	• Glacial lake • GLOFs • GLOF Inventory • Cryosphere
Keywords (Place)	• High Mountain Asia •
Use limitation	Free to use with attribution to the source. Suggested citation: ICIMOD. (2023). GLOF database of High Mountain Asia [Data set]. ICIMOD. https://doi.org/10.26066/RDS.1973283
Access constraints	License
Use constraints	License
Other constraints	Creative Commons Attribution 4.0 International (CC BY 4.0) (https://creativecommons.org/licenses/by/4.0/)
Spatial representation type	Text, table
Metadata language	eng
Character set	UTF8
Topic category	• Geoscientific information

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Distributor ICIMOD - 977-1-5275222	
Hierarchy level	Dataset
Source	•
File identifier	8881454b-6f7c-461b-95c2-eaf7618230d9 XML
Metadata language	eng
Character set	UTF8
Hierarchy level	Attribute
Date stamp	2025-12-15T19:08:14
Metadata standard name	North American Profile of ISO 19115:2003
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Overviews



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